# Andre Boufama

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## Education

Cornell University, B.S. Mechanical Engineering (Minor: Aerospace Engineering) -- *GPA: 3.9* *Expected May 2027*

## Skills

- CAD 3D design: Fusion 360 (6 yrs), Blender (4 yrs), Altium (1 yr) + Aerospace: VTOL UAV and Quadcopter build experience
- Programming: Python (5 yrs), C/C++ (3 yrs), JavaScript, ReactJS (2 yrs), Docker (2 yrs), API frameworks (3 yrs)

## Personal Projects (Photos):

**Finger and Motion Tracking XR Glove**

- Designed and manufactured in one day a gauntlet which tracked the position of two fingers and the rotation of the hand
- Created an immersive computer UI that would support easy calibration and added apps to help with physical therapy

**VTOL Drone Payload Delivery Platform**

- Designed and manufactured 2m wingspan tilt rotor VTOL in using airfoil specifications based on system requirements

**Laparoscopic Surgery Training Device**

- Lead engineer at the Cornell Laparoscopic Surgery club designing and manufacturing training tools for non invasive surgery
- Designed and manufactured a novel foldable laparoscopic simulator under the guidance of organization founder Dr. James Rosser

## Positions

**Full Team Lead & Mechanical Lead at Cornell Physical Intelligence** *August 2025 - present*

- Developed two quadcopter research platforms, led team of engineers to to further the design with custom PCBs and FEA
- Developed unique morphologies and control algorithms, collaborated with startups to develop perception and autonomy

**GeoDesic Lab Researcher** *September 2025 - present*

- Designed PCB for switching motor inputs from a Raspberry Pi to a standard controller on the fly to test novel control algorithms

**LoRaWAN Robotics Swarm Control Research, University of New Hampshire** *June 2022 - August 2023*

- Designed and implemented a LoRa based swarm communication prototype for drones and ground robots; presented poster at New Hampshire EPSCoR Summer STEM Symposium and at IEEE MIT Undergraduate Research Technology Conference

**Data Engineer Intern at Taylor Fresh Foods** *July 2024 - May 2025*

- Programmed Microsoft Powerapps functions integrated with Databricks to optimize energy efficiency and collect environmental data for more efficient warehouse energy partitioning
- Trained multiplatform genAI RAG for internal indexing using Databricks data lakehouse and deployed company wide for internal testing, presented for senior leadership

**AI Summer Camp Counselor, EPIC robotic programming camp** *July 2021 - August 2025*

- Designed coding and AI focused curriculum and robotics activities for middle schoolers; led classroom instruction and supervised five camp volunteers and 200+ campers, Trained 6DOF robot arms using Nvidia Groot N1 foundation model

**Camp Counselor, Camp Horizons** *June 2025 - August 2025*

- Supported 24/7 campers of all ages with developmental disabilities to provide a classic summer camp experience

## Awards

- U.S. Presidential Scholar
- Modular Hydroponics System, won **First Prize** Award in Autodesk “Green Future Student Design Challenge” (https://aboufama.github.io/website/), Top 50 Autodesk Instructables of 2024
- Overpass Oasis Suspension Bridge, **First Prize** Award, Autodesk “Make it Bridge” Design Challenge (https://aboufama.github.io/website/)

## **Two of my favorite projects** (Full portfolio: [https://aboufama.github.io/website/](https://aboufama.github.io/website/))

**Autonomous All Terrain Rover -** *Unique traversal geometry platform for environmental data collection*

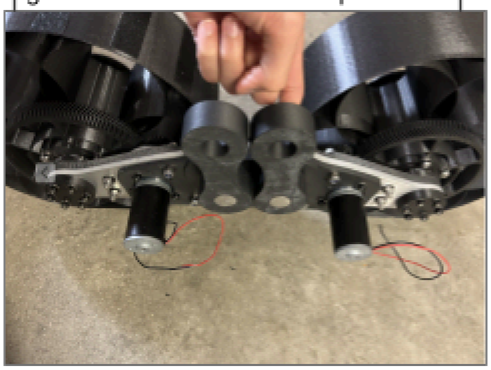
1:3 scale rover to test the kinematic system (disconnected rocker bogie system)

I designed, manufactured, and programmed a **heavy lift autonomous all terrain rover**. Conducted research and took inspiration from URC (University Rover Challenge) for my basic design.

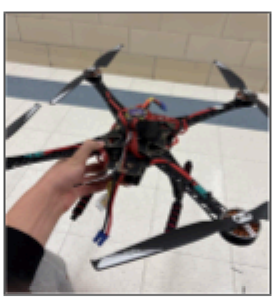
I mocked up a few potential configurations, and 3D printed scale models (see left image) to see how the wheels would interact with rough terrain.

Later, testing was conducted in the field to verify the 3D printed parts and the extrusion bolted body were able to withstand up to 40 lb of payload.

Custom 3D printed herringbone motor gearboxes for increased torque



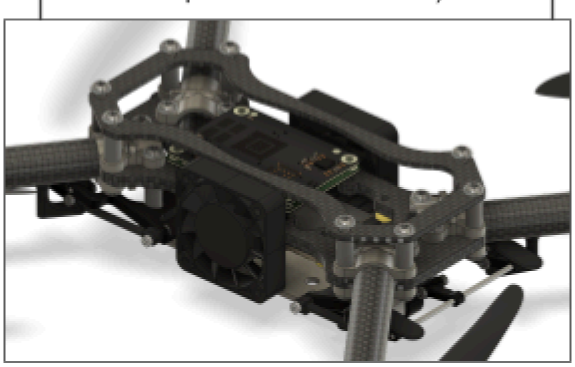
**Experimental Quadcopter UAV Design** - *Novel direct control architecture and ultra lightweight assembly*



As part of a new engineering club at my college, I'm designing a **drone platform** to test a novel control method which eliminates the commercial flight controller.

I first put together a proof of concept (see left), and am still drafting the final drone design. The image shown below is the V4 overhaul, which includes a novel way of mounting the arms so that you can use a much smaller base plate while maintaining rigidity by using the brackets themselves as support for the plate.

Detailed internal view showing the Raspberry Pi used for direct UART motor control. (Plastic cover holding fans and some other parts has been hidden)



Version 4 of the custom frame design. Layered carbon fiber combined with custom clamps to get the weight under 200g (13 inch props). Added landing legs which fold into the carbon fiber tubing when retracted and lock upon deployment.